

ECON602 TA Notes: Search and Matching in Macro Labor

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This note presents the details of Mortensen and Pissarides (1994, ReStud), the workhorse model of search-and-matching frictions in the macro-labor literature.

1 Model Setup

- There is a continuum $i \in [0, 1]$ of firms in the economy.
- Each firm has one job that can be in one of two states: filled and producing, or vacant and searching. Jobs that are neither producing nor associated with an active search are destroyed.
- Productivity of each job a_{it} is idiosyncratic, with a structure

$$a_{it} = p_t + \sigma \epsilon_{it} \tag{1}$$

where p_t is the aggregate productivity that is common to all jobs, and ϵ_{it} is the job-specific component of productivity. The dispersion parameter σ is common to all jobs as well.

- A Poisson process governs variation in ϵ_{it} . From time t to the next instant $t + \Delta t$ (the model is written in continuous time), *epsilon* remains unchanged with probability $1 - \lambda$, while a new value is drawn from a fixed distribution $F(\cdot)$ with probability λ . The distribution has support $(-\infty, \epsilon_u]$, mean zero, and unit variance.
- Assume that the newly created job has productivity $p + \sigma \epsilon_u$, i.e. the newly created jobs are the most productive ones in the economy (justification?).
- (Search/matching Friction) There is a fixed cost of posting and maintaining vacancy for each job, denoted as c , common to all firms/jobs.
 - Hence, firms are not willing to separate when a negative ϵ shock hits the firm because posting a new vacancy is costly.
 - However, if the negative ϵ shock is large enough, so that ϵ is below some threshold ϵ_d , the job is destroyed. The threshold ϵ_d will be pinned down endogenously in the model.
 - Hence, the job destruction rate is equal to $\lambda F(\epsilon_d)$.
 - In this model, there is no on-the-job search or job-to-job flows, and therefore job destruction = layoff = separation (with job-to-job flows, separation = layoff + job-to-job flow).

- (Matching Technology) Let u denote unemployment and v denote vacancies. The constant-returns-to-scale matching function $m(v, u)$ gives the rate at which vacant jobs and unemployed workers meet, fill jobs, and begin production.
 - In other words, $m(v, u)$ is the job-creation rate in the model. It gives the total number of matches.
 - The job-finding rate is the probability that an unemployed worker finds a job. In this model, all new hires come from unemployment (there are no job-to-job flows), so the job-finding rate is $m(v, u)/u$.
 - Similarly, the job-filling rate is the probability that a vacancy is filled. In the model, it is $m(v, u)/v$. The paper denotes it by q , a function of u/v :

$$q(v/u) \equiv m(v, u)/v = m(1, u/v)$$

with q being a decreasing function.

- The vacancy-to-unemployment ratio v/u is referred to as “tightness”. If the labor market is tight, v/u is large (i.e. there are more jobs than workers, or, a shortage of labor supply in the market), so that the probability of a job being filled q is smaller, and the job finding rate $m(v, u)/u = m(v/u, 1)$ becomes larger.
- The goal is to solve for equilibrium vacancies v , unemployment u , and the job-destruction rate $\lambda F(\epsilon_d)$.

2 Bellman Equations in the Steady State

We first look into the case where there is no aggregate uncertainty, i.e. p does not change over time.

In the paper, the model is written in continuous time. I will explain the intuitions in discrete time and transform it into continuous time. For a full discussion of the continuous time methods, see my note for Pablo’s part on ELMS.

2.1 Value of Vacancy

Let V denote the value to the firm of posting a vacancy, and let $J(\epsilon)$ denote the value of an existing job with idiosyncratic productivity ϵ . In the next period,

- With probability q (job filling rate), the vacancy will be filled, and, with our assumption that the newly created job has productivity ϵ_u , the value of this newly filled job is $J(\epsilon_u)$.
- With probability $1 - q$, the vacancy remains unfilled. Also, posting and maintaining a vacancy costs c units of output for the firms.

As a result, in discrete time, the value of a vacancy is

$$V_t = -c + \frac{1}{1+r} [qJ(\epsilon_u) + (1-q)V_{t+1}] \quad (2)$$

From Discrete-time to Continuous-time

To move from the discrete-time Bellman equation to continuous time, first change the period length from 1 to Δt :

$$V_t = -c\Delta t + \frac{1}{1+r\Delta t} [q\Delta t J(\epsilon_u) + (1-q\Delta t)V_{t+\Delta t}]$$

Rearrange terms:

$$V_t - \frac{1-q\Delta t}{1+r\Delta t} V_{t+\Delta t} = -c\Delta t + \frac{1}{1+r\Delta t} q\Delta t J(\epsilon_u)$$

Divide both sides by Δt :

$$\frac{V_t - \frac{(1-q\Delta t)V_{t+\Delta t}}{1+r\Delta t}}{\Delta t} = -c + \frac{1}{1+r\Delta t} qJ(\epsilon_u)$$

Take the limit $\Delta t \rightarrow 0$: the left-hand side becomes

$$\begin{aligned} \frac{V_t - \frac{(1-q\Delta t)V_{t+\Delta t}}{1+r\Delta t}}{\Delta t} &= \frac{V_t - V_{t+\Delta t}}{\Delta t} + \frac{V_{t+\Delta t} - \frac{1-q\Delta t}{1+r\Delta t} V_{t+\Delta t}}{\Delta t} \\ &= \frac{V_t - V_{t+\Delta t}}{\Delta t} + \frac{(r+q)\Delta t}{1+r\Delta t} \frac{1}{\Delta t} V_{t+\Delta t} \\ &\rightarrow -\dot{V}_t + (r+q)V_t \end{aligned} \quad (3)$$

where $\dot{V}_t \equiv \frac{dV_t}{dt}$. If we mute the aggregate uncertainty as in the paper (so that p does not change over time), $V_t = V_{t+\Delta t} = V$, and hence $\dot{V}_t \equiv \frac{dV_t}{dt} = 0$.

The limit of the right-hand side is $-c + qJ(\epsilon_u)$. Putting everything together:

$$rV = -c + q(J(\epsilon_u) - V) \quad (4)$$

which is exactly equation (1) in the paper.

“Free entry” Condition

In the steady state, the equilibrium value of the vacancy should be 0:

$$rV = 0 \tag{5}$$

If this is not the case, then there are incentives for the firms to enter and create more jobs, driving up the job creation rate. This is akin to the free entry condition that we discussed in Hopenhayn (1992) last semester.

2.2 Wage Determination

Let $W(\epsilon)$ denote the value to a worker of a job with productivity ϵ . Let U denote the worker's reservation value from unemployment (e.g., the value of government unemployment benefits).

Assume that the firm and worker determine a wage contract through cooperative Nash bargaining, so the resulting contract assigns a fixed proportion of total surplus,

$$S(\epsilon) \equiv J(\epsilon) + W(\epsilon) - U$$

to the worker:

$$\begin{cases} W(\epsilon) - U &= \beta S(\epsilon) \\ J(\epsilon) &= (1 - \beta) S(\epsilon) \end{cases} \tag{6}$$

Let $w(\epsilon)$ be the wage that the firm pays the worker to implement this allocation result.

2.3 Value of Existing Jobs

We now write down the Bellman equation for $J(\epsilon)$ and $W(\epsilon)$, the value of job for firms and workers. Again, I explain the intuition in discrete time and transform it into continuous time.

- First, notice that the concurrent payoff of the firm is equal to

$$p + \sigma\epsilon - w(\epsilon)$$

- In the next period, with probability λ , a new $\epsilon = \epsilon'$ is drawn, so that the continuation value in this case is equal to $J(\epsilon')$ if the job survives in the next period. However, if the newly drawn ϵ' is too low, the value of the job may be below 0, so that the job

will be destroyed. Hence, the continuation value when there is a new draw in the next period is $\max\{J(\epsilon'), 0\}$.

- With probability $1 - \lambda$, the productivity ϵ does not change.

Putting everything together, the Bellman equation in discrete time is

$$J(\epsilon) = p + \sigma\epsilon - w(\epsilon) + \frac{1}{1+r} \int_{-\infty}^{\epsilon_u} [\lambda \max\{J(\epsilon'), 0\} + (1-\lambda)J(\epsilon)] dF(\epsilon')$$

Apply the same steps in Section 2.1 to rewrite the above Bellman equation in time interval Δt , rearrange terms and take limit, we get the continuous-time Bellman equation

$$rJ(\epsilon) = p + \sigma\epsilon - w(\epsilon) + \lambda \int_{-\infty}^{\epsilon_u} [\max\{J(\epsilon'), 0\} - J(\epsilon)] dF(\epsilon') \quad (7)$$

Replacing the J 's on the right-hand side using the fact $J(\epsilon) = (1 - \beta)S(\epsilon)$, we get equation (5) in the paper:

$$rJ(\epsilon) = p + \sigma\epsilon - w(\epsilon) + \lambda(1 - \beta) \int_{-\infty}^{\epsilon_u} [\max\{S(\epsilon'), 0\} - S(\epsilon)] dF(\epsilon') \quad (8)$$

Similarly, for the worker, the value of job $W(\epsilon)$ can be written in discrete time as

$$W(\epsilon) = w(\epsilon) + \frac{1}{1+r} \int_{-\infty}^{\epsilon_u} [\lambda \max\{W(\epsilon'), U\} + (1-\lambda)W(\epsilon)] dF(\epsilon')$$

the first term in the integrand on the right-hand side says that if the new draw ϵ' makes the value of job lower than the worker's reservation utility, then the worker would quit the job and become unemployed.

Again, use the procedures in section 2.1 and use the fact that $W(\epsilon) - U = \beta S(\epsilon)$, we get the Bellman equation in continuous time

$$rW(\epsilon) = w(\epsilon) + \lambda \int_{-\infty}^{\epsilon_u} [\max\{W(\epsilon'), U\} - W(\epsilon)] dF(\epsilon') \quad (9)$$

2.4 Value of Unemployment

Lastly we define the value of unemployment. In the next period, the probability of an employed worker getting a job is equal to the job finding rate $m(v, u)/u = qv/u$. Recall that all new hires are with the highest productivity $p + \sigma\epsilon_u$. Hence, the Bellman equation for unemployment is

$$U = b + \frac{1}{1+r} \left[\frac{qv}{u} W(\epsilon_u) + \left(1 - \frac{qv}{u}\right) U \right]$$

which, in continuous time, becomes

$$rU = b + \frac{qv}{u} (W(\epsilon_u) - U) = b + \frac{qv}{u} \beta S(\epsilon_u) \quad (10)$$

where b is the exogenous unemployment benefit that the worker gets from the government.

2.5 Equilibrium Tightness v/u and Destruction Threshold ϵ_d

(8) + (9) - (10):

$$(r + \lambda)S(\epsilon) = p + \sigma\epsilon - b + \lambda \int_{-\infty}^{\epsilon_u} \max\{S(\epsilon'), 0\} dF(\epsilon') - \beta \frac{vq}{u} S(\epsilon_u) \quad (11)$$

The threshold ϵ_d is defined by the point at which the firm is indifferent between shutting down the job and continuing to operate it:

$$J(\epsilon_d) = 0 \quad (12)$$

which is equivalent to $(1 - \beta)S(\epsilon_d) = 0$ and hence $S(\epsilon_d) = 0$. Use this fact to simplify the integral on the right-hand side of (11):

$$\begin{aligned}
\int_{-\infty}^{\epsilon_u} \max\{S(\epsilon'), 0\} dF(\epsilon') &= \int_{\epsilon_d}^{\epsilon_u} S(\epsilon') dF(\epsilon') \\
&= [S(\epsilon')F(\epsilon')]_{\epsilon_d}^{\epsilon_u} - \int_{\epsilon_d}^{\epsilon_u} F(\epsilon')S'(\epsilon') d\epsilon' \\
&= S(\epsilon_u) - 0 - \int_{\epsilon_d}^{\epsilon_u} F(\epsilon')S'(\epsilon') d\epsilon' \\
&= S(\epsilon_u) - S(\epsilon_d) - \int_{\epsilon_d}^{\epsilon_u} F(\epsilon')S'(\epsilon') d\epsilon' \\
&= \int_{\epsilon_d}^{\epsilon_u} S'(\epsilon') d\epsilon' - \int_{\epsilon_d}^{\epsilon_u} F(\epsilon')S'(\epsilon') d\epsilon' \\
&= \int_{\epsilon_d}^{\epsilon_u} (1 - F(\epsilon'))S'(\epsilon') d\epsilon' \\
&= \frac{\sigma}{r + \lambda} \int_{\epsilon_d}^{\epsilon_u} [1 - F(\epsilon')] d\epsilon' \tag{13}
\end{aligned}$$

where

- the second equality follows from integration by parts;
- the third equality follows from

$$S(\epsilon_u)F(\epsilon_u) - S(\epsilon_d)F(\epsilon_d) = S(\epsilon) \cdot 1 - 0 \cdot F(\epsilon_d) = S(\epsilon_u) - 0$$

- the fourth equality replaces the 0 by $S(\epsilon_d)$;
- the last equality follows from the envelope condition of $S(\epsilon)$ in (11):

$$(r + \lambda)S'(\epsilon) = \sigma \implies S'(\epsilon) = \frac{\sigma}{r + \lambda}$$

Substitute (13) back into (11):

$$(r + \lambda)S(\epsilon) = p + \sigma\epsilon - b + \frac{\sigma\lambda}{r + \lambda} \int_{\epsilon_d}^{\epsilon_u} [1 - F(\epsilon')] d\epsilon' - \beta \frac{vq}{u} S(\epsilon_u) \quad (14)$$

This can be further simplified since we know what $S(\epsilon_u)$ is, from the Bellman equation for V (4) and the “free-entry” condition (5):

$$J(\epsilon_u) = \frac{c}{q} \implies S(\epsilon_u) = \frac{1}{1 - \beta} J(\epsilon_u) = \frac{1}{1 - \beta} \frac{c}{q}$$

Substituting this expression into (14) gives

$$(r + \lambda)S(\epsilon) = p + \sigma\epsilon - b + \frac{\sigma\lambda}{r + \lambda} \int_{\epsilon_d}^{\epsilon_u} [1 - F(\epsilon')] d\epsilon' - \frac{\beta c}{1 - \beta} \frac{v}{u} \quad (15)$$

This equation (15) is the key equation for analysis.

Job Destruction Curve: Given v/u , what is ϵ_d ?

The first equation that pins down the equilibrium is the job-destruction curve. For a given tightness v/u , it determines the exit threshold ϵ_d . This result follows from (15) by substituting $\epsilon = \epsilon_d$ on the left-hand side and using the fact that $S(\epsilon_d) = 0$:

$$p + \sigma\epsilon_d = b + \frac{\beta c}{1 - \beta} \frac{v}{u} - \frac{\sigma\lambda}{r + \lambda} \int_{\epsilon_d}^{\epsilon_u} [1 - F(\epsilon')] d\epsilon' \quad (16)$$

Note that the right-hand side is also a function of ϵ_d . Hence, the threshold ϵ is the solution to equation (16).

Note:

- The sum of first two terms on the right-hand side of (16) is the opportunity cost of employment, or equivalently, the value of unemployment (it is essentially the right-hand side of the value function for unemployment (10)).
- The third term is an “option value” for the firm. From our derivation, it is essentially the term $\frac{1-\beta}{1+r} \lambda \int \max\{S(\epsilon'), 0\} dF(\epsilon')$ in the discrete-time version, which is the expected continuation value of a new productivity draw for the current match.

The Shape of JD curve

We are interested in the shape of the job destruction curve. By implicit function theorem and fundamental theorem of calculus, (16) implies that

$$\sigma \frac{\partial \epsilon_d}{\partial (v/u)} = \frac{\beta c}{1 - \beta} + \frac{\sigma \lambda}{r + \lambda} (1 - F(\epsilon_d)) \frac{\partial \epsilon_d}{\partial (v/u)}$$

so that

$$\sigma \left[1 - \frac{\lambda}{r + \lambda} (1 - F(\epsilon_d)) \right] \frac{\partial \epsilon_d}{\partial (v/u)} = \frac{\beta c}{1 - \beta} \implies \frac{\partial \epsilon_d}{\partial (v/u)} > 0$$

Hence, it is upward sloping.

We are also interested in some comparative statics of parameters, holding v/u constant. For instance, we can investigate how ϵ_d varies with $p - b$:

$$1 + \sigma \frac{\partial \epsilon_d}{\partial (p - b)} = \frac{\sigma \lambda}{r + \lambda} (1 - F(\epsilon_d)) \frac{\partial \epsilon_d}{\partial (p - b)} \implies \frac{\partial \epsilon_d}{\partial (p - b)} < 0$$

In the problem set, you will work on more questions of this type. I will not go further here.

Job Creation Curve: given ϵ_d , what is v/u ?

Again, by equation (15),

$$(r + \lambda)S(\epsilon_u) = p + \sigma \epsilon_u - b + \frac{\sigma \lambda}{r + \lambda} \int_{\epsilon_d}^{\epsilon_u} [1 - F(\epsilon')] d\epsilon' - \frac{\beta c}{1 - \beta} \frac{v}{u}$$

$$(r + \lambda)S(\epsilon_d) = p + \sigma \epsilon_d - b + \frac{\sigma \lambda}{r + \lambda} \int_{\epsilon_d}^{\epsilon_u} [1 - F(\epsilon')] d\epsilon' - \frac{\beta c}{1 - \beta} \frac{v}{u}$$

Taking the difference gives

$$(r + \lambda)(S(\epsilon_u) - S(\epsilon_d)) = \sigma(\epsilon_u - \epsilon_d) \tag{17}$$

Using $S(\epsilon_u) = \frac{c}{q(1-\beta)}$ and $S(\epsilon_d) = 0$, we obtain

$$q \left(\frac{v}{u} \right) = \frac{c(r + \lambda)}{(1 - \beta)\sigma(\epsilon_u - \epsilon_d)} \tag{18}$$

Equation (18) is the job creation curve.

It is easy to see that v/u is a decreasing function of ϵ_d . The intersection of the JD curve

(16) and the JC curve (18) pins down the equilibrium tightness $(v/u)^*$ and threshold ϵ_d^* .

2.6 Equilibrium Unemployment and Beveridge Curve

Finally, we determine equilibrium unemployment u . We are effectively working with a heterogeneous-worker model with two types of workers: employed and unemployed. The steady state requires the distribution across these types to remain invariant over time.

We know that the mass of employed and unemployed workers is $1 - u$ and u , respectively. At the steady state, the outflow and inflow into the unemployment should be exactly the same, so that there is zero net entrance/exit into the unemployment. The mass of inflow into unemployment comes from the separation of the existing job match, which is equal to the mass of employment $1 - u$ times the job destruction rate $\lambda F(\epsilon_d^*)$:

$$(1 - u)\lambda F(\epsilon_d^*)$$

and the mass of outflow of unemployment comes from the job finding of the currently unemployed population, which is equal to the mass of unemployment u times the job finding rate $m(v, u)/u = m(v/u, 1)$:

$$u \cdot m((v/u)^*, 1)$$

In a stationary equilibrium, the inflow should coincide with outflow:

$$(1 - u)\lambda F(\epsilon_d^*) = u \cdot m((v/u)^*, 1) \implies u^* = \frac{\lambda F(\epsilon_d^*)}{\lambda F(\epsilon_d^*) + m((v/u)^*, 1)} \quad (19)$$

Equation (19) gives us the equilibrium relationship between vacancy v and unemployment u . It is referred to as the Beveridge Curve in the literature.

2.7 Exercise: A Positive MIT Shock to p

Consider how the Mortensen–Pissarides economy responds to a one-time unanticipated increase in aggregate productivity p .

We first think about the job creation and job destruction curve. From (18) we can see that p does not directly shift the job creation curve. For the job destruction curve, since we have already established earlier that for each fixed tightness v/u , an increase in p will lead to a decrease in ϵ_d . Hence, the JD curve would shift inwards in Figure 1 of the paper, resulting in a higher equilibrium tightness $(v/u)^*$ and a lower job destruction threshold ϵ_d^* .

We then use the Beveridge curve (19) to pin down the response of unemployment and

vacancy. Since ϵ_d^* becomes smaller, the Beveridge curve as in Figure 2 of the paper would shift inwards. Since the equilibrium tightness v^*/u^* gets higher, the JC line in Figure 2 shifts upwards. Hence, we can see that the equilibrium unemployment goes down unambiguously while the direction of equilibrium vacancy is inconclusive.

In sum, as p goes up,

- Job creation rate = Layoff rate = Separation Rate = $\lambda F(\epsilon_d)$ goes down;
- Tightness v/u goes up;
- Unemployment u goes down (countercyclical!);
- Vacancy v is ambiguous.

These objects can be directly computed and mapped to the data counterparts, if one has high-quality employer-employee matched dataset (e.g. LEHD in Census).

3 Anticipated Cyclical Shock in p

What if the firms are able to anticipate the cyclical shock in p ?

Suppose there are in total two possible aggregate states: p or p^* , where $p < p^*$. The Poisson probability of transitioning to the other state is equal to μ .

The paper does not provide the details of deriving equations (16)–(18), so I present the derivations here.

3.1 Value function when current state is low

The equations in the anticipated shock case are largely the same as the steady state equations. Let $J(\epsilon)$ and $J^*(\epsilon)$ denote the value of job to firms when the current aggregate productivity is p and p^* respectively. In discrete time, the firm's value function in the low state is

$$\begin{aligned}
 J(\epsilon) = & p + \sigma\epsilon - w(\epsilon) + \frac{1}{1+r} \left[(1-\mu) \int_{-\infty}^{\epsilon_u} [\lambda \max\{J(\epsilon'), 0\} + (1-\lambda)J(\epsilon)] dF(\epsilon') \right. \\
 & \left. + \mu \int_{-\infty}^{\epsilon_u} [\lambda \max\{J^*(\epsilon'), 0\} + (1-\lambda)J^*(\epsilon)] dF(\epsilon') \right]
 \end{aligned} \tag{20}$$

As before, we transform this equation into the version with time interval being Δt :

$$J(\epsilon) = (p + \sigma\epsilon - w(\epsilon))\Delta t + \frac{1}{1 + r\Delta t} \left[(1 - \mu\Delta t) \int_{-\infty}^{\epsilon_u} [\lambda\Delta t \max\{J(\epsilon'), 0\} + (1 - \lambda\Delta t)J(\epsilon)] dF(\epsilon') \right. \\ \left. + \mu\Delta t \int_{-\infty}^{\epsilon_u} [\lambda\Delta t \max\{J^*(\epsilon'), 0\} + (1 - \lambda\Delta t)J^*(\epsilon)] dF(\epsilon') \right] \quad (21)$$

Again, move the $J(\epsilon)$ term to the left, and divide both sides by Δt :

$$\frac{1}{\Delta t} \left(J(\epsilon) - \frac{(1 - \lambda\Delta t)(1 - \mu\Delta t)}{1 + r\Delta t} J(\epsilon) \right) \\ = p + \sigma\epsilon - w(\epsilon) + \frac{1}{1 + r\Delta t} \left[(1 - \mu\Delta t) \int_{-\infty}^{\epsilon_u} \lambda \max\{J(\epsilon'), 0\} dF(\epsilon') \right. \\ \left. + \mu \int_{-\infty}^{\epsilon_u} [\lambda\Delta t \max\{J^*(\epsilon'), 0\} + (1 - \lambda\Delta t)J^*(\epsilon)] dF(\epsilon') \right]$$

We take the limit $\Delta t \rightarrow 0$ on both sides. The left-hand side becomes

$$\text{LHS} = \frac{1}{\Delta t} \frac{r\Delta t + \mu\Delta t + \lambda\Delta t + \lambda\mu(\Delta t)^2}{1 + r\Delta t} J(\epsilon) \rightarrow \frac{r + \lambda + \mu + 2\lambda\mu\Delta t}{1 + 2r\Delta t} J(\epsilon) \rightarrow (r + \lambda + \mu)J(\epsilon)$$

and the right-hand side becomes

$$\text{RHS} \rightarrow p + \sigma\epsilon - w(\epsilon) + \int_{-\infty}^{\epsilon_u} \lambda \max\{J(\epsilon'), 0\} dF(\epsilon') + \mu J^*(\epsilon)$$

where the penultimate step follows from L'Hôpital's rule. Hence, we obtain the continuous-time version of the value function J :

$$(r + \lambda + \mu)J(\epsilon) = p + \sigma\epsilon - w(\epsilon) + \lambda \int_{-\infty}^{\epsilon_u} \max\{J(\epsilon'), 0\} dF(\epsilon') + \mu J^*(\epsilon) \quad (22)$$

Similarly, we can write down the worker's value function

$$(r + \lambda + \mu)W(\epsilon) = w(\epsilon) + \lambda \int_{-\infty}^{\epsilon_u} \max\{W(\epsilon'), U\} dF(\epsilon') + \mu W^*(\epsilon) \quad (23)$$

You may derive equation (23) yourself as an exercise.

The value of unemployment in discrete time is

$$U = b + \frac{1}{1+r} \left[\frac{qv}{u} [(1-\mu)W(\epsilon_u) + \mu W^*(\epsilon_u)] + \left(1 - \frac{qv}{u}\right) [(1-\mu)U + \mu U^*] \right] \quad (24)$$

and change the length of period to Δt :

$$U = b\Delta t + \frac{1}{1+r\Delta t} \left[\frac{qv}{u} \Delta t [(1-\mu\Delta t)W(\epsilon_u) + \mu\Delta t W^*(\epsilon_u)] + \left(1 - \frac{qv}{u}\Delta t\right) [(1-\mu\Delta t)U + \mu\Delta t U^*] \right]$$

and hence we get the continuous time Bellman equation

$$rU = b + \frac{qv}{u} (W(\epsilon_u) - U) + \mu(U^* - U) \quad (25)$$

Combine (22), (23) and (25), and use the fact that

$$W(\epsilon_u) - U = \beta S(\epsilon_u) \text{ and } W^*(\epsilon_u) - U = \beta S^*(\epsilon_u)$$

we can get a continuous-time Bellman equation for the joint surplus $S(\epsilon) = J(\epsilon) + W(\epsilon) - U$:

$$\begin{aligned} r(J(\epsilon) + W(\epsilon) - U) &= p + \sigma\epsilon - b \\ &+ \lambda \int_{\epsilon_d}^{\epsilon_u} J(\epsilon') + \max\{W(\epsilon'), U\} dF(\epsilon') - \lambda U + \lambda U - \lambda J(\epsilon) - \lambda W(\epsilon) \\ &+ \mu(J^*(\epsilon) + W^*(\epsilon) - U^*) + \mu U - \frac{qv}{u} (W(\epsilon_u) - U) \\ &- \mu J(\epsilon) - \mu W(\epsilon) \end{aligned}$$

Hence

$$rS(\epsilon) = p + \sigma\epsilon - b + \lambda \int_{\epsilon_d}^{\epsilon_u} S(\epsilon') dF(\epsilon') - (\lambda + \mu)S(\epsilon) + \mu S^*(\epsilon) - \beta \frac{vq}{u} S(\epsilon_u)$$

and therefore

$$(r + \lambda + \mu)S(\epsilon) = p + \sigma\epsilon - b + \lambda \int_{\epsilon_d}^{\epsilon_u} S(\epsilon') dF(\epsilon') - \beta \frac{vq}{u} S(\epsilon_u) + \mu S^*(\epsilon) \quad (26)$$

Equation (26) here is Equation (16) in the paper.

3.2 Value functions when current state is high

When the current state is p^* , the value functions can be a bit tricky.

Again, we think about the value of the job for the firm $J^*(\epsilon)$. The firm's profit in the current period remains $p^* + \sigma\epsilon - w^*(\epsilon)$. In the next period, with probability $1 - \mu$ the aggregate state remains p^* , and hence with probability $\lambda(1 - \mu)$ the firm would draw a new idiosyncratic shock ϵ' under p^* from distribution F , and with $(1 - \mu)(1 - \lambda)$ probability the idiosyncratic state remains the same under p^* . These parts are identical to the previous analysis.

What really makes things tricky is the contingencies where the aggregate productivity transitions into the low state p in the next period: since we know from the MIT shock exercise that we should expect the separation threshold to be greater in the low state, i.e. $\epsilon_d > \epsilon_d^*$. This means that if $\epsilon \in [\epsilon_d^*, \epsilon_d]$, the job would disappear in the next period, which gives the firm value 0. Hence, if $\epsilon \in [\epsilon_d^*, \epsilon_d]$, we have

$$J^*(\epsilon) = p^* + \sigma\epsilon - w^*(\epsilon) + \frac{1}{1+r} \left\{ (1-\mu) \int_{-\infty}^{\epsilon_u} [\lambda \max\{J^*(\epsilon'), 0\} + (1-\lambda)J^*(\epsilon)] dF(\epsilon') \right. \\ \left. + \mu \left[(1-\lambda) \cdot 0 + \lambda \int_{-\infty}^{\epsilon_u} \max\{J(\epsilon'), 0\} dF(\epsilon') \right] \right\} \quad (27)$$

which results in the continuous-time value function

$$(r + \lambda + \mu)J^*(\epsilon) = p^* + \sigma\epsilon - w^*(\epsilon) + \lambda \int_{\epsilon_d^*}^{\epsilon_u} \underbrace{J^*(\epsilon')}_{=(1-\beta)S^*(\epsilon')} dF(\epsilon') \quad (28)$$

Similarly, when $\epsilon \in [\epsilon_d^*, \epsilon_d]$, value of job for the worker is equal to

$$W^*(\epsilon) = w^*(\epsilon) + \frac{1}{1+r} \left[(1-\mu) \int_{-\infty}^{\epsilon_u} \lambda [\max\{W^*(\epsilon'), U^*\} + (1-\lambda)W^*(\epsilon)] dF(\epsilon') + \mu U \right] \quad (29)$$

which gives us

$$(r + \lambda + \mu)W^*(\epsilon) = w^*(\epsilon) + \lambda \int_{-\infty}^{\epsilon_u} \max\{W^*(\epsilon'), U^*\} dF(\epsilon') + \mu U = w^*(\epsilon) + \lambda \int_{\epsilon_d^*}^{\epsilon_u} \beta S^*(\epsilon') dF(\epsilon') + \lambda U^* + \mu U \quad (30)$$

The value of employment in the high state, U^* , is similar to (25), except for replacing the

variables with superscript *:

$$rU^* = b + \frac{q^*v^*}{u^*}(W^*(\epsilon_u) - U^*) + \mu(U - U^*) \quad (31)$$

Hence

$$\begin{aligned} rS^*(\epsilon) &= r(J^*(\epsilon) + W^*(\epsilon) - U^*) \\ &= p^* + \sigma\epsilon - b + \lambda \int_{\epsilon_d^*}^{\epsilon_u} S^*(\epsilon') dF(\epsilon') - (\lambda + \mu)J^*(\epsilon) - (\lambda + \mu)W^*(\epsilon) + \lambda U^* + \mu U \\ &\quad - \beta \frac{q^*v^*}{u^*} S^*(\epsilon_u) - \mu U + \mu U^* \\ &= p^* + \sigma\epsilon - b + \lambda \int_{\epsilon_d^*}^{\epsilon_u} S^*(\epsilon') dF(\epsilon') - (\lambda + \mu)S^*(\epsilon) - \beta \frac{q^*v^*}{u^*} S^*(\epsilon_u) \end{aligned} \quad (32)$$

This is equation (18) in the paper.

When $\epsilon > \epsilon_d$, then the firm will keep the job as the aggregate state decreases from p^* to p . I write down the discrete-time Bellman equation: firm's value

$$\begin{aligned} J^*(\epsilon) &= p^* + \sigma\epsilon - w^*(\epsilon) + \frac{1}{1+r} \left[(1-\mu) \int_{-\infty}^{\epsilon_u} [\lambda \max\{J^*(\epsilon'), 0\} + (1-\lambda)J^*(\epsilon)] dF(\epsilon') \right. \\ &\quad \left. + \mu \int_{-\infty}^{\epsilon_u} [\lambda \max\{J(\epsilon'), 0\} + (1-\lambda)J(\epsilon)] dF(\epsilon') \right] \end{aligned} \quad (33)$$

worker's value

$$\begin{aligned} W^*(\epsilon) &= w^*(\epsilon) + \frac{1}{1+r} \left[(1-\mu) \int_{-\infty}^{\epsilon_u} [\lambda \max\{W^*(\epsilon'), U^*\} + (1-\lambda)W^*(\epsilon)] dF(\epsilon') \right. \\ &\quad \left. + \mu \int_{-\infty}^{\epsilon_u} [\lambda \max\{W(\epsilon'), U\} + (1-\lambda)W(\epsilon)] dF(\epsilon') \right] \end{aligned} \quad (34)$$

and unemployment value same as above. We end up with a continuous time Bellman equation for S^* :

$$(r + \lambda + \mu)S^*(\epsilon) = p^* + \sigma\epsilon - b + \lambda \int_{\epsilon_d}^{\epsilon_u} S^*(\epsilon') dF(\epsilon') - \beta \frac{v^*q^*}{u^*} S^*(\epsilon_u) + \mu S(\epsilon) \quad (35)$$

which is equation (17) of the paper.

3.3 Implication: Asymmetry in Booms and Busts

The cool part of this exercise is that we get asymmetry for boom and recessions:

- In recessions, there is a “cleansing effect” for the jobs with productivity between $[\epsilon_d^*, \epsilon_d]$: these jobs with an “intermediate” level of idiosyncratic productivity exist when the aggregate state is high but will be destroyed immediately when the aggregate state is low. In the model, this is the source of massive, abrupt layoffs in recessions (although the importance of the cleansing effect is questionable empirically since J2J flows seem to be important, according to JH’s work).
- In booms, however, these jobs $[\epsilon_d^*, \epsilon_d]$ are not created immediately, because of the search and matching friction (captured by the cost of posting vacancy c). Hence, the model has the potential to capture slow recovery of employment in the recovery period.
- Very beautiful work, consistent with the empirics in many aspects.